ICT2217 Network Security

Lab Exercise 1.7: Address Spoofing Attacks and Defense

Learning Outcomes

Upon completion of this laboratory exercise, you should be able to:

- Conduct and defend against IP and MAC address spoofing attacks

Required Hardware

- 1 x Rack of Cisco networking devices
- 1 x Box of cables containing
  - USB-to-RJ45 console cables
  - Ethernet cables
- 3 x Laptops

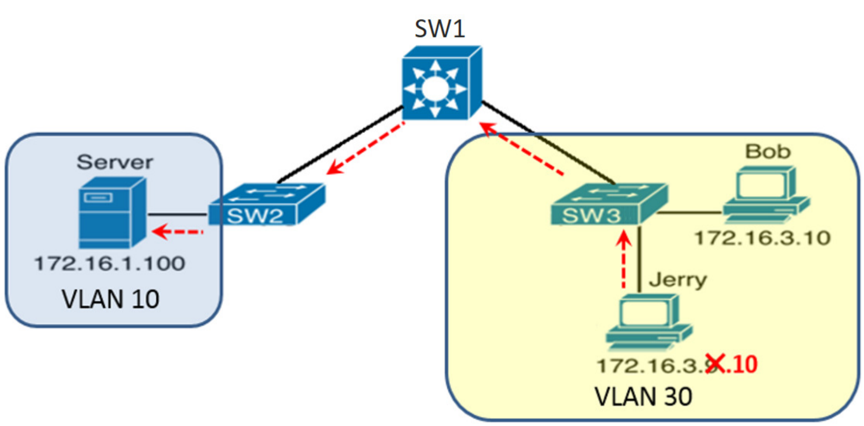
Required Software

- Tera Term https://teratermproject.github.io/index-en.html
- Kali Linux Live Boot on USB drive

SCENARIO A:

PART 1: Setting up Network with ACL to Filter Inter-VLAN Routing Traffic

1.1 Set up the network topology as shown:



1.2 Configure switch SW1 with suitable IP addresses and enable inter-VLAN routing.

1.3 Boot up Server, Bob and Jerry's PC in Kali Linux and configure them with suitable default gateway, subnet mask and IP addresses as shown before IP address spoofing attack.

1.4 Verify that both Bob and Jerry are able to ping the Server to ensure your setup is correct.

1.5 Next, implement ACL on SW1 to only permit Bob to visit the Server.

1.6 When done, verify that Bob is still able to ping the Server but not Jerry.

1.7 Return to SW1, show its arp cache.

```
SW1# show arp
```

What are the cache entries for Bob and Jerry?

ARP cache before attack	
Internet Address	Physical Address

PART 2: Conducting IP Address Spoofing Attack

2.1 Suppress Kali from sending out ARP probe which will prevent it from using conflicting IP address:

```
(kali@kali)$ sudo ip link set dev eth0 arp off
```

2.2 Configure IP address of Jerry's PC to that of Bob and verify that it has been changed successfully.

2.3 Turn ARP back on again to allow normal network operations:

```
(kali@kali)$ sudo ip link set dev eth0 arp on
```

2.4 Now, is Jerry able to ping the Server?

2.5 At SW1, show its arp cache again. Did you observe any difference as compared to Step 1.7?

ARP cache after attack	
Internet Address	Physical Address

PART 3: Defending against IP Address Spoofing Attack using IP Source Guard

3.1 Implement IPSG (IP Source Guard) against IP address spoofing as discussed.

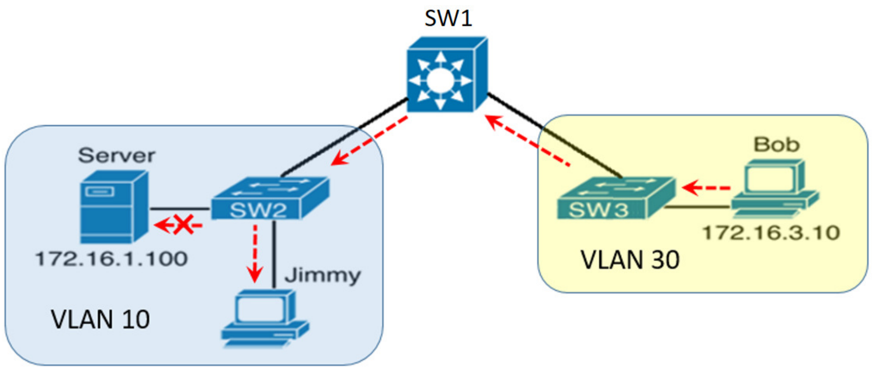
3.2 When done, ping from Jerry's PC to the Server again. What do you observe?

3.3 Do you think IPSG is an effective defense against IP address spoofing attack?

SCENARIO B:

PART 4: Conducting IP and MAC Address Spoofing Attack

4.1 Modify the network topology in Part 1 to a different scenario as shown:



4.2 Verify that Bob is able to ping the Server successfully.

4.3 Observe the MAC address table at switch SW2. What is the interface associated with the MAC address of Server?

MAC address table before attack	
Address	Interface

4.4 Boot up Jimmy's PC in Kali and spoof its IP address to that of the Server.

4.5 In addition, start up a terminal at Jimmy's PC and enter the following commands (use colon hexadecimal format *xx:xx:xx:xx:xx:xx* for MAC address) to spoof its MAC address to Server's MAC address:

```
(kali@kali)$ sudo su
(root@kali)# ifconfig eth0 hw ether spoof-mac-address
```

4.6 Verify that both the IP and MAC addresses of Jimmy's PC have been changed successfully.

4.7 From Jimmy's PC, ping an unused IP address, e.g. 172.16.1.10 to update the MAC address table of switch SW2.

4.8 Observe the MAC address table at switch SW2 again. What is the interface associated with the MAC address of Server now?

MAC address table after attack	
Address	Interface

4.9 Start Wireshark at the Server and Jimmy’s PC.

4.10 From host Bob, ping the Server IP address.

4.11 Stop Wireshark at the Server and Jimmy’s PC and try to find any captured ping packets sent by Bob.

4.12 Where did Bob's ping packets go to? Why?

PART 5: Defending against IP and MAC Address Spoofing Attacks using IP Source Guard

5.1 Implement IPSG (IP Source Guard) to defend against IP and MAC address spoofing as discussed.

5.2 When done, repeat Part 4 again and observe what happens.

5.3 Do you think IPSG is an effective defense against IP and MAC address spoofing attacks?